

CHAIN OF IDEAS: REVOLUTIONIZING RESEARCH VIA NOVEL IDEA DEVELOPMENT WITH LLM AGENTS

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ABSTRACT

Effective research ideation is a critical step for scientific research. However, the exponential increase in scientific literature makes it challenging for researchers to stay current with recent advances and identify meaningful research directions. Recent developments in large language models (LLMs) suggest a promising avenue for automating the generation of novel research ideas. However, existing methods for idea generation either trivially prompt LLMs or directly expose LLMs to extensive literature without indicating useful information. Inspired by the research process of human researchers, we propose a Chain-of-Ideas (CoI) agent, an LLM-based agent that organizes relevant literature in a chain structure to effectively mirror the progressive development in a research domain. This organization facilitates LLMs to capture the current advancements in research, thereby enhancing their ideation capabilities. Furthermore, we propose Idea Arena, an evaluation protocol that can comprehensively evaluate idea generation methods from different perspectives, aligning closely with the preferences of human researchers. Experimental results indicate that the CoI agent consistently outperforms other methods and shows comparable quality as humans in research idea generation. Moreover, our CoI agent is budget-friendly, with a minimum cost of \$0.50 to generate a candidate idea and its corresponding experimental design¹.

1 INTRODUCTION

Idea generation is a crucial aspect of scientific research for driving technological innovations and breakthroughs. Traditionally, this process has been predominantly human-driven, necessitating expert researchers to review extensive literature, identify limitations in existing solutions, and propose new research directions. However, the complexity and vastness of scientific literature, coupled with rapid technological advancements, have rendered this task increasingly challenging for researchers.

Recent advancements in large language models (LLMs) (Achiam et al., 2023; Dubey et al., 2024; Yang et al., 2024a) have enabled these models to exceed human experts in various scientific tasks, including mathematics (Yu et al., 2023), theorem proving (Yang et al., 2023), and coding (Chen et al., 2021). Building on this robust scientific foundation, one may hypothesize that LLMs could support a more abstract and creative research idea-generation task. Notably, Si et al. (2024); Kumar

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¹The code is released at <https://github.com/DAMO-NLP-SG/CoI-Agent>. Try our online demo at https://huggingface.co/spaces/DAMO-NLP-SG/CoI_Agent.

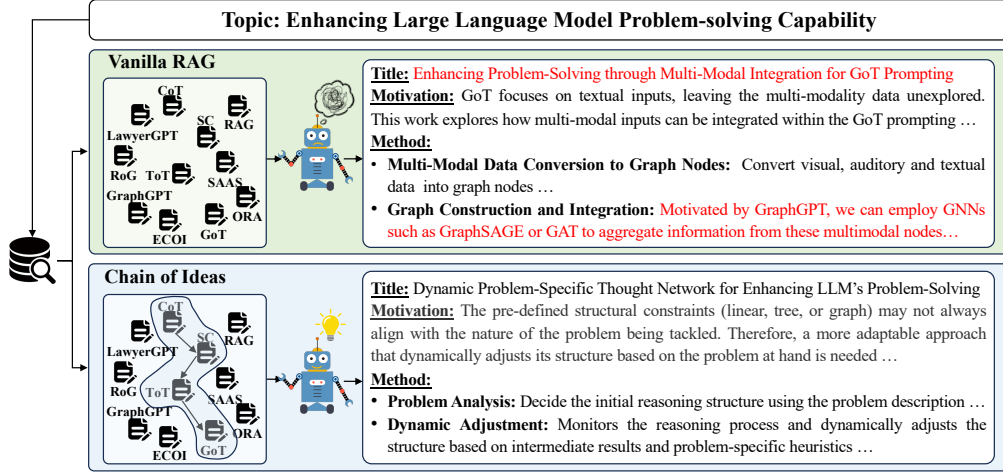


Figure 1: Comparison between the vanilla retrieval augmented generation (RAG) research agent and our Chain-of-Ideas agent on the idea generation task.

et al. (2024) have validated this hypothesis, highlighting its substantial potential to expedite the discovery of novel concepts and uncharted research avenues.

Existing methods seek to address two key challenges to improve the quality of generated ideas: curating pertinent literature for LLMs to gain inspiration and ensuring the novelty of generated ideas. To address the first challenge, previous research enhances traditional academic retrieval systems, which typically depend on textual similarity, with academic knowledge graphs (Baek et al., 2024; Wang et al., 2023). For the second challenge, existing approaches either apply predefined criteria such as novelty to guide the idea generation process (Baek et al., 2024) or iteratively refine ideas until they demonstrate low embedding similarities with existing papers (Wang et al., 2023).

However, in existing attempts, LLMs are presented with an extensive volume of research literature when asked to generate ideas. This makes LLMs vulnerable to the influence of less relevant works, potentially resulting in ideas that lack logical coherence and technological innovation. As shown in the upper part of Figure 1, the LLM borrows an idea from GraphGPT (Tang et al., 2024) and applies it into GoT framework (Besta et al., 2024) to generate what they interpret as a “novel idea”. However, the resultant idea conflates two concepts: GoT is a prompting method while GraphGPT is a fine-tuning method leveraging graph neural network architecture (Zhou et al., 2020). In contrast, human researchers often trace the evolution of a research field by analyzing its progression from foundational works to the most recent advancements. This comprehensive perspective provides valuable insights into the key factors driving developments within the domain. Such an understanding enables researchers to critically assess the limitations of earlier studies while identifying emerging trends. Therefore, they are better grounded in devising innovative and impactful research ideas.

Motivated by the human practices in conducting research, we introduce a novel Chain-of-Ideas (CoI) agent framework to address the previously identified logical inconsistencies in the ideation processes of LLMs. As shown in the bottom part of Figure 1, CoI agent aims to provide a clear landscape of current research topics by systematically selecting and organizing the relevant papers and their ideas in a chain structure. CoI agent offers several distinctive advantages: Firstly, it minimizes the risk of interference from less relevant literature via carefully selecting papers (i.e. from CoT (Wei et al., 2022) to GoT). Second, LLMs are demonstrated with human practice to craft a novel idea. For example, SC (Wang et al., 2022) emerges as a novel idea derived from CoT. This can be viewed as a form of few-shot prompting strategy, which has been proven to enhance the overall LLM’s generation capability (Brown et al., 2020). Third, CoI exemplifies a global progression in research development. As a result, LLMs can gain a deep understanding of the motivations behind these developmental trends, facilitating the identification of promising future research directions.

Specifically, CoI agent first retrieves an anchor paper of the given research topic. Instead of indiscriminately aggregating all papers within the citation network of the anchor, as done in (Baek et al., 2024), we construct the CoI by selecting relevant and important literature from both the anchor’s references and its subsequent works, thereby extending the chain backward and forward from the

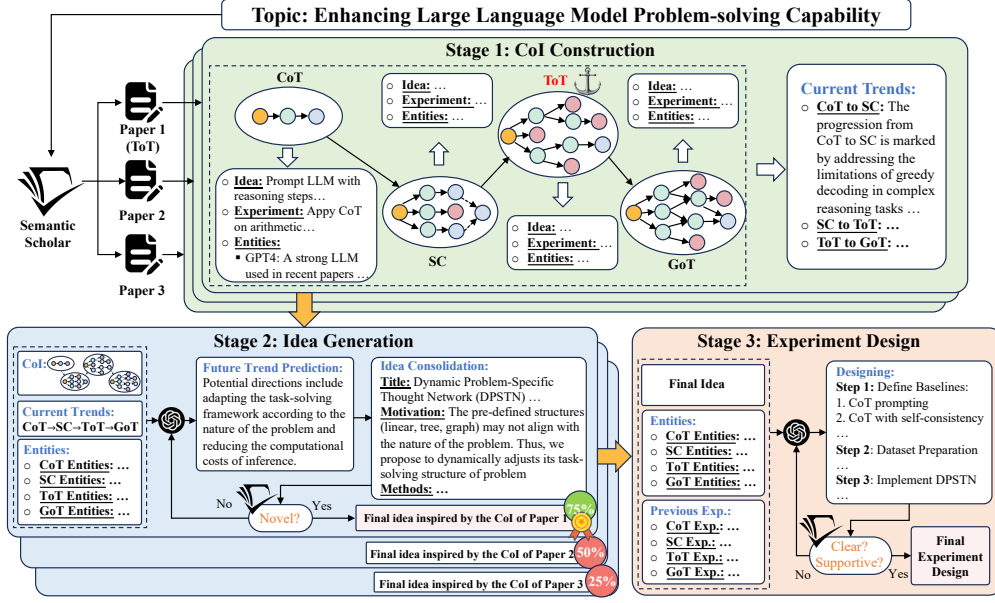


Figure 2: Our proposed CoI agent framework.

anchor. We then prompt the constructed CoI to an LLM for idea generation and experiment design. During idea generation, we require the LLM to predict possible future trends. This prognostic result facilitates the gradual consolidation of the idea, beginning with the motivation for the proposed idea, progressing through an assessment of its potential impact, and culminating in the realization. However, as the evolution of scientific discovery can emerge from multiple perspectives, a single CoI may be insufficient to capture the most promising direction. Additionally, there is no guarantee that the generated ideas will be novel. To address these issues, we construct multiple CoI branches for different perspectives of a research topic. Additionally, a novelty-checker agent iteratively evaluates the draft idea against existing literature and refines it if substantial similarity is identified.

We compare our CoI agent against existing baselines on idea generation in the artificial intelligence (AI) field. To do this, we develop an arena-style evaluation framework called Idea Arena where participant methods compete in pairs, which demonstrates high agreement with human evaluation. The experimental results show that CoI agent consistently ranks first among all automated baselines, surpassing the second-best one by 56 ELO scores in human evaluation. CoI agent can generate ideas as novel as those of human experts. Our analysis further shows that for LLMs to generate novel ideas, a clear developmental trend analysis is more pivotal than the quantity of related literature.

Our contributions are summarized as follows: 1) We propose the CoI agent to enhance LLMs’ capability in idea generation. CoI agent organizes relevant literature in a chain structure to effectively mirror the progressive nature of research development, allowing LLMs to better grasp the current research advancements. 2) We propose Idea Arena for a comprehensive evaluation of idea-generation methods, which shows high agreement with human researchers. 3) Extensive experiments demonstrate the effectiveness of our CoI agent in generating ideas that are comparable to human creativity.

2 METHOD

2.1 FRAMEWORK: CHAIN-OF-IDEAS AGENT

In this section, we detail our CoI agent framework, as illustrated in Figure 2, which consists of three stages: (1) CoI Construction, (2) Idea Generation, and (3) Experiment Design. First, given a research topic, the CoI agent constructs multiple CoIs from existing literature, reflecting different trends within the domain. Then, for each CoI, the LLM predicts future research directions, and crafts ideas through step-by-step consolidation and iterative novelty checks. The best idea is then selected. Lastly, the LLM generates and refines an experiment design to implement the final idea.